

Trajectory Uncertainty Calibration for Multimodal VLMs

Complete Benchmark & Transfer LaTeX Table Compendium

M3-LLaVA (7B) & MQT-LLaVA (7B) Evaluations

September 25, 2026

Abstract

This document compiles all 37 publication-ready LaTeX tables across Groups 1–8 and RQ1 Master Tables for local inspection before Overleaf integration. All tables follow the standardized 4-metric schema: ECE (%) ↓, Ada-ECE (%) ↓, Brier score ↓, and AUROC ↑, featuring Option A ranking (**bold** Rank 1, *italic* Rank 2), Core 7 macro aggregation, Trajectory Platt (5D) evaluation, dedicated VQAv2 single-pass vs. multi-rollout comparison, and isolated RQ1 Core 7 and Adversarial master benchmarks.

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Table 1: **Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)**. Pairwise zero-shot calibration transfer from source datasets across held-out target vision-language benchmarks without target domain supervision. Evaluated across Expected Calibration Error (**ECE** ↓), Adaptive ECE (**Ada-ECE** ↓), AUROC (↑), and Brier score (↓). **Bold**: Rank 1; *italic*: Rank 2 among transfer methods.

Source: VQAv2 (General Scene VQA)												
Method	DocVQA (Document Text)				ChartQA (Diagrams)				TextVQA (Scene Text)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0187	0.0181	0.888	0.0377	0.0331	0.0288	0.818	0.0752	0.0578	0.0535	0.835	0.1628
Uncalibrated (NC)	0.7400	0.7400	<i>0.797</i>	0.6348	0.7143	0.7143	0.799	0.6214	<i>0.0851</i>	<i>0.0867</i>	0.837	0.1717
Temperature Scaling (TS)	<i>0.6968</i>	<i>0.6968</i>	<i>0.797</i>	<i>0.5567</i>	<i>0.6710</i>	<i>0.6710</i>	0.799	<i>0.5490</i>	0.1030	0.1012	0.837	<i>0.1739</i>
Platt Scaling (1D)	0.7562	0.7562	<i>0.797</i>	0.6240	0.7242	0.7242	0.799	0.6094	0.1788	0.1785	0.837	0.2066
Platt 5D (Ours)	0.5865	0.5865	0.808	0.4247	0.5912	0.5912	<i>0.767</i>	0.4566	0.0736	0.0754	<i>0.811</i>	0.1793
Source: VizWiz-VQA (Assistive / Blurry)												
Method	ChartQA (Diagrams)				DocVQA (Document Text)				TextVQA (Scene Text)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0331	0.0288	0.818	0.0752	0.0187	0.0181	0.888	0.0377	0.0578	0.0535	0.835	0.1628
Uncalibrated (NC)	0.7143	0.7143	0.799	0.6214	0.7400	0.7400	<i>0.797</i>	0.6348	0.0851	0.0867	0.837	0.1717
Temperature Scaling (TS)	0.6873	0.6873	0.799	0.5753	0.7134	0.7134	<i>0.797</i>	0.5854	0.0886	<i>0.0851</i>	0.837	0.1717
Platt Scaling (1D)	<i>0.6112</i>	<i>0.6112</i>	0.799	<i>0.5092</i>	<i>0.6321</i>	<i>0.6321</i>	<i>0.797</i>	<i>0.5153</i>	<i>0.0842</i>	0.0896	0.837	<i>0.1689</i>
Platt 5D (Ours)	0.6079	0.6079	0.799	0.5081	0.6263	0.6263	0.799	0.5122	0.0776	0.0779	<i>0.835</i>	0.1683
Source: TextVQA (Scene Text)												
Method	ChartQA (Diagrams)				DocVQA (Document Text)				VizWiz-VQA (Assistive)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0331	0.0288	0.818	0.0752	0.0187	0.0181	0.888	0.0377	0.0638	0.0657	0.794	0.1828
Uncalibrated (NC)	0.7143	0.7143	<i>0.799</i>	0.6214	0.7400	0.7400	<i>0.797</i>	0.6348	0.1703	0.1703	0.789	0.2118
Temperature Scaling (TS)	0.7054	0.7054	<i>0.799</i>	0.6058	0.7314	0.7314	<i>0.797</i>	0.6182	0.1660	0.1660	0.789	0.2102
Platt Scaling (1D)	<i>0.6688</i>	<i>0.6688</i>	<i>0.799</i>	<i>0.5716</i>	<i>0.6921</i>	<i>0.6921</i>	<i>0.797</i>	<i>0.5822</i>	0.0885	0.0873	0.789	0.1881
Platt 5D (Ours)	0.6560	0.6560	0.801	0.5561	0.6647	0.6647	0.809	0.5500	<i>0.1192</i>	<i>0.1192</i>	<i>0.780</i>	<i>0.1988</i>

1 Main Manuscript Tables (Groups 5, 2, 6, 7, 4, 9)

1.1 Table 2: Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)

Table 2: **Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)**. Pairwise zero-shot calibration transfer from source datasets across held-out target vision-language benchmarks without target domain supervision. Evaluated across Expected Calibration Error (**ECE** ↓), Adaptive ECE (**Ada-ECE** ↓), AUROC (↑), and Brier score (↓). **Bold**: Rank 1; *italic*: Rank 2 among transfer methods.

Source: VQAv2 (General Scene VQA)												
Method	DocVQA (Document Text)				ChartQA (Diagrams)				TextVQA (Scene Text)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0108	0.0206	0.594	0.0364	0.0244	0.0392	0.603	0.0765	0.0525	0.0776	0.719	0.1948
Uncalibrated (NC)	0.5324	0.5324	0.592	0.4143	0.7055	0.7055	<i>0.404</i>	0.6483	0.3168	0.3168	<i>0.722</i>	0.3096
Temperature Scaling (TS)	<i>0.5174</i>	<i>0.5174</i>	0.592	<i>0.3377</i>	<i>0.6172</i>	<i>0.6172</i>	<i>0.404</i>	<i>0.4958</i>	<i>0.2757</i>	<i>0.2757</i>	<i>0.722</i>	<i>0.2753</i>
Platt Scaling (1D)	0.6405	0.6405	0.592	0.4583	0.6838	0.6838	<i>0.404</i>	0.5583	0.3714	0.3714	<i>0.722</i>	0.3388
Platt 5D (Ours)	0.4570	0.4570	<i>0.578</i>	0.2880	0.5814	0.5802	0.563	0.4368	0.2356	0.2357	0.730	0.2569
Source: VizWiz-VQA (Assistive / Blurry)												
Method	ChartQA (Diagrams)				DocVQA (Document Text)				TextVQA (Scene Text)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0244	0.0392	0.603	0.0765	0.0108	0.0206	0.594	0.0364	0.0525	0.0776	0.719	0.1948
Uncalibrated (NC)	0.7055	0.7055	<i>0.404</i>	0.6483	0.5324	0.5324	<i>0.592</i>	0.4143	0.3168	0.3168	<i>0.722</i>	0.3096
Temperature Scaling (TS)	0.5836	0.5836	<i>0.404</i>	0.4449	0.5087	0.5087	<i>0.592</i>	0.3172	0.2565	0.2565	<i>0.722</i>	0.2654
Platt Scaling (1D)	<i>0.4165</i>	<i>0.4174</i>	<i>0.404</i>	<i>0.3500</i>	<i>0.2477</i>	<i>0.2477</i>	<i>0.592</i>	<i>0.1660</i>	<i>0.1475</i>	<i>0.1447</i>	<i>0.722</i>	<i>0.2331</i>
Platt 5D (Ours)	0.3709	0.3702	0.423	0.2973	0.2055	0.2055	0.603	0.1350	0.1340	0.1316	0.736	0.2185
Source: TextVQA (Scene Text)												
Method	ChartQA (Diagrams)				DocVQA (Document Text)				VizWiz-VQA (Assistive)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0244	0.0392	0.603	0.0765	0.0108	0.0206	0.594	0.0364	0.0603	0.0817	0.795	0.1418
Uncalibrated (NC)	0.7055	0.7055	<i>0.404</i>	0.6483	0.5324	0.5324	<i>0.592</i>	0.4143	0.3203	0.3203	0.809	0.2488
Temperature Scaling (TS)	0.4622	0.4622	<i>0.404</i>	0.2919	0.4743	0.4743	<i>0.592</i>	0.2630	0.2855	0.2926	0.809	0.2435
Platt Scaling (1D)	<i>0.3122</i>	<i>0.3122</i>	<i>0.404</i>	<i>0.1979</i>	<i>0.2571</i>	<i>0.2571</i>	<i>0.592</i>	<i>0.1174</i>	<i>0.1028</i>	<i>0.1179</i>	0.809	<i>0.1521</i>
Platt 5D (Ours)	0.2849	0.2849	0.517	0.1792	0.2118	0.2118	0.608	0.1001	0.0868	0.0842	<i>0.765</i>	0.1501

1.2 Table 3: Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)

Table 3: **Macro-Averaged Calibration Benchmark Across Core 7 Datasets (M3-LLaVA 7B).** **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	34.98	34.83	0.3475	<i>0.782</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	21.42	21.86	0.2168	<i>0.782</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.27	5.69	<i>0.1442</i>	<i>0.782</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>5.42</i>	<i>5.78</i>	0.1425	0.800

Table 4: **Macro-Averaged Calibration Benchmark Across Core 7 Datasets (MQT-LLaVA 7B).** **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	35.55	35.37	0.3364	0.662
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	25.15	25.24	0.2308	0.662
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>6.12</i>	<i>7.45</i>	<i>0.1478</i>	<i>0.665</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	4.96	5.97	0.1431	0.708

1.3 Group 2: Macro-Averaged Calibration Benchmark Across Core 7 Datasets

Table 5: **Comprehensive Calibration Benchmark Breakdown Across Core 7 Datasets.** Evaluated across Adaptive ECE (Ada-ECE (%) ↓) and AUROC (↑) on M3-LLaVA (7B) (Panel A) and MQT-LLaVA (7B) (Panel B). **Bold**: best; *italic*: second best within each metric column. Clean unshaded presentation.

Calibration Method	AI2D		ChartQA		DocVQA		ScienceQA		TextVQA		VizWiz-VQA		VQAv2		Average	
	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑
Panel A: M3-LLaVA (7B)																
Naive Confidence (NC)	42.00	<i>0.614</i>	71.03	<i>0.807</i>	74.57	<i>0.845</i>	21.93	<i>0.807</i>	11.07	0.839	15.97	<i>0.793</i>	7.23	<i>0.772</i>	34.83	<i>0.782</i>
Temperature Scaling (TS)	12.18	<i>0.614</i>	44.47	<i>0.807</i>	48.89	<i>0.845</i>	12.68	<i>0.807</i>	11.14	0.839	15.09	<i>0.793</i>	8.58	<i>0.772</i>	21.86	<i>0.782</i>
Platt Scaling (1D)	7.70	<i>0.614</i>	<i>3.62</i>	<i>0.807</i>	<i>2.52</i>	<i>0.845</i>	4.99	<i>0.807</i>	<i>8.22</i>	0.839	5.80	<i>0.793</i>	<i>6.97</i>	<i>0.772</i>	5.69	<i>0.782</i>
Trajectory Platt (5D)	<i>11.15</i>	0.661	2.88	0.818	1.81	0.888	<i>6.15</i>	0.813	5.35	<i>0.835</i>	<i>6.57</i>	0.794	6.50	0.788	<i>5.78</i>	0.800
Panel B: MQT-LLaVA (7B)																
Naive Confidence (NC)	26.92	0.746	71.30	0.491	53.30	<i>0.517</i>	23.17	<i>0.760</i>	31.91	<i>0.684</i>	30.66	0.797	<i>10.31</i>	<i>0.641</i>	35.37	0.662
Temperature Scaling (TS)	13.10	0.746	44.88	0.491	47.07	<i>0.517</i>	10.23	<i>0.760</i>	18.32	<i>0.684</i>	29.41	0.797	13.65	<i>0.641</i>	25.24	0.662
Platt Scaling (1D)	<i>8.25</i>	0.746	<i>4.27</i>	<i>0.509</i>	<i>3.67</i>	<i>0.517</i>	<i>6.91</i>	<i>0.760</i>	<i>10.46</i>	<i>0.684</i>	7.62	0.797	10.94	<i>0.641</i>	<i>7.45</i>	<i>0.665</i>
Trajectory Platt (5D)	6.53	<i>0.739</i>	3.92	0.603	2.06	0.594	5.72	0.769	7.76	0.719	<i>8.17</i>	<i>0.795</i>	7.64	0.739	5.97	0.708

1.4 Comprehensive Core 7 Dataset Benchmark Breakdown (M3 & MQT 7B)

Table 6: **RQ1 Evaluation on M3-LLaVA (7B)**. Comparison of calibration methods across seven vision-language benchmarks. Metrics: Expected Calibration Error (ECE (%) ↓), Adaptive ECE (Ada-ECE (%) ↓), Brier Score (↓), and AUROC (↑). **Bold**: Rank 1; *italic*: Rank 2 among single-pass methods.

Dataset	Naive Confidence				Temperature Scaling				Platt Scaling (1D)				Trajectory Platt (5D)			
	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC
AI2D	42.00	42.00	0.4200	<i>0.614</i>	8.62	12.18	0.2477	<i>0.614</i>	5.63	7.70	<i>0.2370</i>	<i>0.614</i>	<i>8.61</i>	<i>11.15</i>	0.2289	0.661
ChartQA	71.03	71.03	0.6151	<i>0.807</i>	44.47	44.47	0.2782	<i>0.807</i>	3.23	<i>3.62</i>	<i>0.0755</i>	<i>0.807</i>	<i>3.31</i>	2.88	0.0752	0.818
DocVQA	74.57	74.57	0.6430	<i>0.845</i>	48.89	48.89	0.2796	<i>0.845</i>	<i>2.62</i>	<i>2.52</i>	<i>0.0405</i>	<i>0.845</i>	1.87	1.81	0.0377	0.888
ScienceQA	22.16	21.93	0.2368	<i>0.807</i>	13.10	12.68	0.1924	<i>0.807</i>	4.80	4.99	<i>0.1732</i>	<i>0.807</i>	<i>5.17</i>	<i>6.15</i>	0.1719	0.813
TextVQA	10.50	11.07	0.1714	0.839	10.38	11.14	0.1712	0.839	<i>8.47</i>	<i>8.22</i>	0.1615	0.839	5.78	5.35	<i>0.1628</i>	<i>0.835</i>
VizWiz-VQA	15.97	15.97	0.2071	<i>0.793</i>	15.09	15.09	<i>0.2065</i>	<i>0.793</i>	<i>6.64</i>	5.80	0.1828	<i>0.793</i>	6.38	<i>6.57</i>	0.1828	0.794
VQAv2	8.64	7.23	<i>0.1387</i>	<i>0.772</i>	9.40	8.58	0.1417	<i>0.772</i>	5.52	<i>6.97</i>	0.1391	<i>0.772</i>	<i>6.86</i>	6.50	0.1380	0.788
Macro Avg.	34.98	34.83	0.3475	<i>0.782</i>	21.42	21.86	0.2168	<i>0.782</i>	5.27	5.69	<i>0.1442</i>	<i>0.782</i>	<i>5.42</i>	<i>5.78</i>	0.1425	0.800

Table 7: **RQ1 Evaluation on MQT-LLaVA (7B)**. Comparison of calibration methods across seven vision-language benchmarks. Metrics: Expected Calibration Error (ECE (%) ↓), Adaptive ECE (Ada-ECE (%) ↓), Brier Score (↓), and AUROC (↑). **Bold**: Rank 1; *italic*: Rank 2 among single-pass methods.

Dataset	Naive Confidence				Temperature Scaling				Platt Scaling (1D)				Trajectory Platt (5D)			
	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC
AI2D	27.15	26.92	0.2887	0.746	13.31	13.10	0.2197	0.746	5.92	<i>8.25</i>	0.2015	0.746	<i>6.27</i>	6.53	<i>0.2035</i>	<i>0.739</i>
ChartQA	71.30	71.30	0.6483	0.491	44.88	44.88	0.2779	0.491	<i>3.06</i>	<i>4.27</i>	0.0762	<i>0.509</i>	2.44	3.92	<i>0.0765</i>	0.603
DocVQA	53.30	53.30	0.4162	<i>0.517</i>	47.07	47.07	0.2584	<i>0.517</i>	0.09	<i>3.67</i>	0.0362	<i>0.517</i>	<i>1.08</i>	2.06	<i>0.0364</i>	0.594
ScienceQA	23.65	23.17	0.2579	<i>0.760</i>	9.32	10.23	0.1975	<i>0.760</i>	<i>7.99</i>	<i>6.91</i>	<i>0.1916</i>	<i>0.760</i>	6.46	5.72	0.1860	0.769
TextVQA	32.27	31.91	0.3286	<i>0.684</i>	18.21	18.32	0.2503	<i>0.684</i>	<i>8.95</i>	<i>10.46</i>	<i>0.2148</i>	<i>0.684</i>	5.25	7.76	0.1948	0.719
VizWiz-VQA	30.66	30.66	0.2435	0.797	29.41	29.41	0.2328	0.797	<i>6.27</i>	7.62	<i>0.1433</i>	0.797	6.03	<i>8.17</i>	0.1418	<i>0.795</i>
VQAv2	<i>10.49</i>	<i>10.31</i>	0.1713	<i>0.641</i>	13.87	13.65	0.1789	<i>0.641</i>	10.54	10.94	<i>0.1712</i>	<i>0.641</i>	7.16	7.64	0.1629	0.739
Macro Avg.	35.55	35.37	0.3364	0.662	25.15	25.24	0.2308	0.662	<i>6.12</i>	<i>7.45</i>	<i>0.1478</i>	<i>0.665</i>	4.96	5.97	0.1431	0.708

1.5 RQ1 Master Table: Core 7 Vision-Language Calibration Benchmark (M3 & MQT 7B)

tabularx array

Table 8: **RQ1 Evaluation: Calibration Robustness Under Adversarial and Safety Stress Benchmarks.** Evaluated across Expected Calibration Error (ECE (%) ↓), Adaptive ECE (Ada-ECE (%) ↓), Brier Score (↓), and AUROC (↑) on M3-LLaVA (7B) (Panel A) and MQT-LLaVA (7B) (Panel B) across AVQA, VLLM-Safety, and Adversarial Macro Average. **Bold:** Rank 1; *italic:* Rank 2 within each metric column among single-pass methods. Multi-pass baselines report placeholders (–) pending rerun completion. Clean unshaded presentation.

Calibration Method	AVQA				VLLM-Safety				Macro Average			
	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Panel A: M3-LLaVA (7B)												
Naive Confidence (NC)	25.66	25.40	0.3194	<i>0.599</i>	<i>11.59</i>	<i>10.50</i>	<i>0.1732</i>	0.814	18.63	17.95	0.2463	<i>0.707</i>
Temperature Scaling (TS)	4.45	17.96	0.2480	<i>0.599</i>	16.98	16.80	0.1812	0.814	<i>10.71</i>	17.38	0.2146	<i>0.707</i>
Platt Scaling (1D)	6.84	<i>17.67</i>	<i>0.2463</i>	<i>0.599</i>	14.76	14.04	0.1798	0.814	10.80	<i>15.85</i>	<i>0.2131</i>	<i>0.707</i>
LN-Entropy	-	-	-	-	-	-	-	-	-	-	-	-
Semantic Entropy	-	-	-	-	-	-	-	-	-	-	-	-
EigenScore	-	-	-	-	-	-	-	-	-	-	-	-
UMPIRE	-	-	-	-	-	-	-	-	-	-	-	-
Trajectory Platt (5D)	<i>5.26</i>	7.45	0.2240	0.691	5.87	6.88	0.1709	<i>0.773</i>	5.57	7.17	0.1975	0.732
Panel B: MQT-LLaVA (7B)												
Naive Confidence (NC)	27.88	28.00	0.3385	<i>0.602</i>	<i>9.63</i>	<i>9.74</i>	0.1966	0.763	18.76	18.87	0.2675	<i>0.682</i>
Temperature Scaling (TS)	6.56	<i>13.28</i>	0.2491	<i>0.602</i>	14.20	14.42	0.2054	0.763	10.38	13.85	0.2272	<i>0.682</i>
Platt Scaling (1D)	1.18	13.66	<i>0.2449</i>	<i>0.602</i>	11.34	11.29	<i>0.1937</i>	0.763	<i>6.26</i>	<i>12.48</i>	<i>0.2193</i>	<i>0.682</i>
LN-Entropy	-	-	-	-	-	-	-	-	-	-	-	-
Semantic Entropy	-	-	-	-	-	-	-	-	-	-	-	-
EigenScore	-	-	-	-	-	-	-	-	-	-	-	-
UMPIRE	-	-	-	-	-	-	-	-	-	-	-	-
Trajectory Platt (5D)	<i>5.02</i>	6.54	0.2332	0.654	5.68	6.58	0.1921	<i>0.733</i>	5.35	6.56	0.2127	0.694

1.6 RQ1 Master Table: Adversarial & Safety Benchmark (M3 & MQT 7B)

Table 9: **VQAv2 Calibration: Single-Pass Trajectory Calibration (1×) vs. Multi-Rollout UMPIRE Suite (10×) (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0, K = 1$)	8.64	7.23	<i>0.1387</i>	0.772
Temperature Scaling (TS)	Single-Pass ($T = 0.0, K = 1$)	9.40	8.58	0.1417	0.772
Platt Scaling (1D)	Single-Pass ($T = 0.0, K = 1$)	5.52	<i>6.97</i>	0.1391	0.772
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	12.40	-	-	0.787
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	11.40	-	-	0.690
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	6.20	-	-	<i>0.816</i>
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	<i>5.80</i>	-	-	0.819
Trajectory Platt (5D)	Single-Pass ($T = 0.0, K = 1$)	6.86	6.50	0.1380	0.788

Table 10: **VQAv2 Calibration: Single-Pass Trajectory Calibration (1×) vs. Multi-Rollout UMPIRE Suite (10×) (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0, K = 1$)	10.49	<i>10.31</i>	0.1713	0.641
Temperature Scaling (TS)	Single-Pass ($T = 0.0, K = 1$)	13.87	13.65	0.1789	0.641
Platt Scaling (1D)	Single-Pass ($T = 0.0, K = 1$)	10.54	10.94	<i>0.1712</i>	0.641
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	12.30	-	-	0.767
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	13.70	-	-	0.699
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	10.20	-	-	<i>0.773</i>
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	<i>7.80</i>	-	-	0.790
Trajectory Platt (5D)	Single-Pass ($T = 0.0, K = 1$)	7.16	7.64	0.1629	0.739

1.7 Group 9: VQAv2 Single-Pass (1×) vs. Multi-Rollout UMPIRE Suite (10×)

Table 11: **Zero-Shot Cross-Domain Source Anchor Leaderboard.** Performance of Trajectory Platt (5D) across candidate source anchors vs. standard transfer baselines across 12 unseen target benchmarks. Evaluated zero-shot on M3-LLaVA (7B) and MQT-LLaVA (7B). **Bold:** Rank 1; *Italic:* Rank 2 among transfer configurations. ↓: lower is better; ↑: higher is better.

Configuration / Source Anchor	M3-LLaVA (7B)				MQT-LLaVA (7B)				Dual-Arch
	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓	Total Wins ↑
IN-DOMAIN CALIBRATION (PLATT 5D)	4.77	5.85	0.720	0.1356	5.53	6.42	0.743	0.1312	—
<i>Zero-Shot Transfer Baselines (Macro Mean Across All Pairs)</i>									
Naive Confidence (NC)	40.76	40.73	0.673	0.4019	34.25	34.20	0.688	0.3230	—
Temperature Scaling (TS)	33.51	34.97	0.673	0.3107	30.07	30.43	0.688	0.2581	—
Platt Scaling (1D)	38.02	38.27	0.642	0.3640	31.39	31.48	0.594	0.2944	—
<i>Trajectory Platt (5D) Candidate Source Anchors</i>									
VQAv2	39.34	39.35	0.679	0.3689	33.81	33.58	0.662	0.3022	22/24
VizWiz-VQA	40.85	40.88	0.662	0.3864	<i>20.94</i>	<i>20.86</i>	<i>0.685</i>	<i>0.2090</i>	<i>20/24</i>
DocVQA	<i>31.66</i>	<i>31.62</i>	<i>0.675</i>	0.3328	37.85	37.87	0.644	0.3684	18/24
TextVQA	41.36	41.35	0.667	0.4002	20.65	20.79	0.668	0.1868	<i>20/24</i>
ChartQA	31.33	31.51	0.662	<i>0.3326</i>	36.06	36.07	0.476	0.3555	16/24
POPE	42.77	42.82	0.668	0.4163	34.04	33.97	0.673	0.3197	9/24

1.8 Group 6: Zero-Shot Source Anchor Leaderboard

Table 12: **Macro-Averaged Calibration Scaling Across Training Budgets (M3-LLaVA 7B).** Evaluated across all 14 benchmarks over 5 subsampling seeds. **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Budget (N)	Calibration Method	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro AUROC ↑	Macro Brier ↓
$N = 50$	Naive Confidence (NC)	41.24	41.23	<i>0.674</i>	0.4023
	Temperature Scaling (TS)	22.42	23.19	<i>0.674</i>	0.2245
	Platt Scaling (1D)	5.64	6.93	0.694	0.1413
	Trajectory Platt (5D)	<i>6.79</i>	<i>7.58</i>	<i>0.674</i>	<i>0.1456</i>
$N = 100$	Naive Confidence (NC)	41.24	41.23	<i>0.674</i>	0.4023
	Temperature Scaling (TS)	22.25	23.11	<i>0.674</i>	0.2244
	Platt Scaling (1D)	5.03	6.42	0.691	0.1399
	Trajectory Platt (5D)	<i>5.89</i>	<i>7.08</i>	0.691	<i>0.1415</i>
$N = 200$	Naive Confidence (NC)	41.24	41.23	0.674	0.4023
	Temperature Scaling (TS)	22.15	23.05	0.674	0.2240
	Platt Scaling (1D)	4.65	6.08	<i>0.700</i>	<i>0.1390</i>
	Trajectory Platt (5D)	<i>5.07</i>	<i>6.49</i>	0.701	0.1385
$N = 500$	Naive Confidence (NC)	41.24	41.23	0.674	0.4023
	Temperature Scaling (TS)	22.10	23.00	0.674	0.2239
	Platt Scaling (1D)	4.08	5.81	<i>0.698</i>	<i>0.1384</i>
	Trajectory Platt (5D)	<i>4.68</i>	<i>5.97</i>	0.713	0.1365
$N = 1000$	Naive Confidence (NC)	41.24	41.23	0.674	0.4023
	Temperature Scaling (TS)	22.09	23.01	0.674	0.2239
	Platt Scaling (1D)	3.89	5.70	<i>0.698</i>	<i>0.1383</i>
	Trajectory Platt (5D)	<i>4.53</i>	<i>5.90</i>	0.718	0.1359
$N = 1500$	Naive Confidence (NC)	41.24	41.23	0.674	0.4023
	Temperature Scaling (TS)	22.13	23.01	0.674	0.2239
	Platt Scaling (1D)	3.84	5.69	<i>0.698</i>	<i>0.1383</i>
	Trajectory Platt (5D)	<i>4.68</i>	<i>5.99</i>	0.720	0.1356
Full	Naive Confidence (NC)	41.24	41.23	0.674	0.4023
	Temperature Scaling (TS)	22.19	23.01	0.674	0.2239
	Platt Scaling (1D)	3.90	5.69	<i>0.698</i>	<i>0.1383</i>
	Trajectory Platt (5D)	<i>4.77</i>	<i>5.85</i>	0.720	0.1356

1.9 Group 7: Sample Efficiency & Low-Data Calibration Scaling

Table 13: **Macro-Averaged Calibration Scaling Across Training Budgets (MQT-LLaVA 7B).** Evaluated across all 14 benchmarks over 5 subsampling seeds. **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Budget (N)	Calibration Method	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro AUROC ↑	Macro Brier ↓
$N = 50$	Naive Confidence (NC)	33.05	32.62	<i>0.673</i>	0.3154
	Temperature Scaling (TS)	24.15	24.46	<i>0.673</i>	0.2184
	Platt Scaling (1D)	6.60	8.13	0.678	0.1370
	Trajectory Platt (5D)	<i>7.41</i>	<i>8.29</i>	0.669	<i>0.1400</i>
$N = 100$	Naive Confidence (NC)	33.05	32.62	0.673	0.3154
	Temperature Scaling (TS)	23.93	24.36	0.673	0.2178
	Platt Scaling (1D)	5.89	<i>7.50</i>	<i>0.678</i>	0.1351
	Trajectory Platt (5D)	<i>6.33</i>	7.14	0.706	<i>0.1359</i>
$N = 200$	Naive Confidence (NC)	33.05	32.62	0.673	0.3154
	Temperature Scaling (TS)	23.78	24.24	0.673	0.2173
	Platt Scaling (1D)	5.53	<i>7.32</i>	<i>0.679</i>	<i>0.1342</i>
	Trajectory Platt (5D)	<i>5.70</i>	6.51	0.707	0.1333
$N = 500$	Naive Confidence (NC)	33.05	32.62	0.673	0.3154
	Temperature Scaling (TS)	23.75	24.27	0.673	0.2175
	Platt Scaling (1D)	5.23	<i>7.22</i>	<i>0.679</i>	<i>0.1340</i>
	Trajectory Platt (5D)	<i>5.30</i>	6.40	0.726	0.1314
$N = 1000$	Naive Confidence (NC)	33.05	32.62	<i>0.673</i>	0.3154
	Temperature Scaling (TS)	23.74	24.20	<i>0.673</i>	0.2173
	Platt Scaling (1D)	5.27	<i>7.13</i>	<i>0.673</i>	<i>0.1337</i>
	Trajectory Platt (5D)	<i>5.50</i>	6.31	0.742	0.1311
$N = 1500$	Naive Confidence (NC)	33.05	32.62	<i>0.673</i>	0.3154
	Temperature Scaling (TS)	23.69	24.19	<i>0.673</i>	0.2172
	Platt Scaling (1D)	5.30	<i>7.15</i>	<i>0.673</i>	<i>0.1337</i>
	Trajectory Platt (5D)	<i>5.46</i>	6.48	0.743	0.1311
Full	Naive Confidence (NC)	33.05	32.62	<i>0.673</i>	0.3154
	Temperature Scaling (TS)	23.64	24.18	<i>0.673</i>	0.2172
	Platt Scaling (1D)	5.36	<i>7.15</i>	<i>0.673</i>	<i>0.1337</i>
	Trajectory Platt (5D)	<i>5.53</i>	6.42	0.743	0.1312

Table 14: **Leave-One-Dataset-Out (LODO) Cross-Domain Transfer (M3-LLaVA 7B)**. Macro-averaged across all 14 held-out target benchmarks. Trained on pooled 13 benchmarks. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Transfer Protocol	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Zero-Shot Base	40.93	40.92	0.4018	0.681
Naive Confidence (NC)	Target Adapted (Saerens-EM)	53.39	53.39	0.5308	0.597
Temperature Scaling (TS)	Zero-Shot Base	<i>25.81</i>	28.25	0.2491	0.681
Temperature Scaling (TS)	Target Adapted (Saerens-EM)	54.48	54.48	0.5447	0.681
Platt Scaling (1D)	Zero-Shot Base	27.36	<i>27.81</i>	<i>0.2479</i>	0.681
Platt Scaling (1D)	Target Adapted (Saerens-EM)	39.49	39.49	0.3949	0.681
Trajectory Platt (5D)	Zero-Shot Base	23.36	23.47	0.2213	<i>0.677</i>
Trajectory Platt (5D)	Target Adapted (Saerens-EM)	28.59	28.63	0.2853	<i>0.677</i>

Table 15: **Leave-One-Dataset-Out (LODO) Cross-Domain Transfer (MQT-LLaVA 7B)**. Macro-averaged across all 14 held-out target benchmarks. Trained on pooled 13 benchmarks. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Transfer Protocol	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Zero-Shot Base	32.16	32.16	0.3127	0.694
Naive Confidence (NC)	Target Adapted (Saerens-EM)	54.42	54.42	0.5213	0.611
Temperature Scaling (TS)	Zero-Shot Base	28.72	29.30	0.2471	0.694
Temperature Scaling (TS)	Target Adapted (Saerens-EM)	58.60	58.60	0.5860	0.694
Platt Scaling (1D)	Zero-Shot Base	23.94	24.06	0.2170	0.694
Platt Scaling (1D)	Target Adapted (Saerens-EM)	28.34	28.32	0.2702	0.694
Trajectory Platt (5D)	Zero-Shot Base	<i>19.99</i>	<i>19.91</i>	<i>0.1962</i>	<i>0.691</i>
Trajectory Platt (5D)	Target Adapted (Saerens-EM)	18.55	18.44	0.1930	<i>0.691</i>

1.10 Group 4: Leave-One-Dataset-Out (LODO) Cross-Domain Transfer

2 Group 3: Temperature Transfer Robustness Ablation

Evaluating calibrators trained at $T = 0.0$ and tested across $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and temperature Mean.

2.1 Temperature Transfer: Macro-Averaged Across 4 Benchmarks

Table 16: **Temperature Transfer Robustness (Macro-Averaged Across 4 Benchmarks) (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	13.02	13.34	0.1759	<i>0.794</i>
Temperature Scaling (TS)	10.71	10.82	<i>0.1646</i>	<i>0.794</i>
Platt Scaling (1D)	<i>6.02</i>	5.83	0.1515	<i>0.794</i>
Trajectory Platt (5D)	5.54	<i>6.17</i>	0.1515	0.798
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	28.46	27.71	0.2767	<i>0.780</i>
Temperature Scaling (TS)	25.61	24.92	0.2486	<i>0.780</i>
Platt Scaling (1D)	<i>23.60</i>	<i>22.85</i>	<i>0.2359</i>	<i>0.780</i>
Trajectory Platt (5D)	22.47	21.83	0.2308	0.784
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	18.97	18.02	0.2045	<i>0.799</i>
Temperature Scaling (TS)	14.84	14.49	0.1815	<i>0.799</i>
Platt Scaling (1D)	<i>11.45</i>	<i>11.47</i>	<i>0.1693</i>	<i>0.799</i>
Trajectory Platt (5D)	11.07	10.51	0.1668	0.803
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	11.89	11.75	0.1582	0.840
Temperature Scaling (TS)	9.54	9.41	0.1458	0.840
Platt Scaling (1D)	5.78	5.20	0.1351	0.840
Trajectory Platt (5D)	<i>7.67</i>	<i>7.01</i>	<i>0.1371</i>	0.840
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	9.92	10.03	<i>0.1426</i>	0.841
Temperature Scaling (TS)	<i>9.25</i>	9.07	0.1390	0.841
Platt Scaling (1D)	9.07	<i>9.12</i>	0.1390	0.841
Trajectory Platt (5D)	10.92	10.68	0.1464	<i>0.840</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	16.45	16.17	0.1916	<i>0.811</i>
Temperature Scaling (TS)	13.99	13.74	0.1759	<i>0.811</i>
Platt Scaling (1D)	11.19	10.89	0.1662	<i>0.811</i>
Trajectory Platt (5D)	<i>11.53</i>	<i>11.24</i>	<i>0.1665</i>	0.813

Table 17: **Temperature Transfer Robustness (Macro-Averaged Across 4 Benchmarks) (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	22.65	22.51	0.2301	<i>0.779</i>
Temperature Scaling (TS)	15.11	15.72	0.1927	<i>0.779</i>
Platt Scaling (1D)	<i>6.79</i>	<i>7.41</i>	<i>0.1598</i>	<i>0.779</i>
Trajectory Platt (5D)	5.32	6.53	0.1532	0.789
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	41.12	40.92	0.3853	<i>0.758</i>
Temperature Scaling (TS)	28.26	28.20	0.2669	<i>0.758</i>
Platt Scaling (1D)	<i>24.89</i>	<i>24.42</i>	<i>0.2500</i>	<i>0.758</i>
Trajectory Platt (5D)	20.66	20.61	0.2198	0.786
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	34.52	34.17	0.3170	<i>0.763</i>
Temperature Scaling (TS)	21.79	21.89	0.2245	<i>0.763</i>
Platt Scaling (1D)	<i>13.50</i>	<i>13.28</i>	<i>0.1855</i>	<i>0.763</i>
Trajectory Platt (5D)	10.48	10.51	0.1633	0.794
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	25.72	25.83	0.2443	<i>0.789</i>
Temperature Scaling (TS)	17.81	18.31	0.1988	<i>0.789</i>
Platt Scaling (1D)	<i>7.14</i>	<i>7.08</i>	<i>0.1529</i>	<i>0.789</i>
Trajectory Platt (5D)	5.03	5.13	0.1393	0.817
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	20.45	20.69	0.2017	<i>0.805</i>
Temperature Scaling (TS)	19.86	19.93	0.1983	<i>0.805</i>
Platt Scaling (1D)	<i>8.72</i>	8.61	<i>0.1507</i>	<i>0.805</i>
Trajectory Platt (5D)	7.79	<i>8.69</i>	0.1439	0.820
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	28.89	28.82	0.2757	<i>0.779</i>
Temperature Scaling (TS)	20.57	20.81	0.2162	<i>0.779</i>
Platt Scaling (1D)	<i>12.21</i>	<i>12.16</i>	<i>0.1798</i>	<i>0.779</i>
Trajectory Platt (5D)	9.86	10.29	0.1639	0.801

2.2 Temperature Transfer: POPE

Table 18: **Temperature Transfer Robustness on pope (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	3.45	4.40	0.0882	<i>0.739</i>
Temperature Scaling (TS)	4.27	<i>4.37</i>	<i>0.0883</i>	<i>0.739</i>
Platt Scaling (1D)	<i>4.19</i>	4.32	0.0882	<i>0.739</i>
Trajectory Platt (5D)	4.82	6.62	0.0887	0.751
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	9.83	8.69	0.0980	<i>0.764</i>
Temperature Scaling (TS)	9.69	<i>8.53</i>	0.0972	<i>0.764</i>
Platt Scaling (1D)	9.69	8.55	0.0972	<i>0.764</i>
Trajectory Platt (5D)	<i>9.71</i>	8.43	<i>0.0975</i>	0.770
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	7.00	6.53	0.0912	<i>0.786</i>
Temperature Scaling (TS)	<i>6.73</i>	6.02	<i>0.0904</i>	<i>0.786</i>
Platt Scaling (1D)	6.64	<i>5.97</i>	<i>0.0904</i>	<i>0.786</i>
Trajectory Platt (5D)	7.45	5.76	0.0902	0.794
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	4.34	4.56	<i>0.0882</i>	<i>0.826</i>
Temperature Scaling (TS)	<i>5.08</i>	4.43	0.0883	<i>0.826</i>
Platt Scaling (1D)	5.16	<i>4.47</i>	0.0883	<i>0.826</i>
Trajectory Platt (5D)	6.77	4.82	0.0872	0.832
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	5.38	6.14	0.0953	<i>0.876</i>
Temperature Scaling (TS)	<i>6.83</i>	<i>6.80</i>	<i>0.0970</i>	<i>0.876</i>
Platt Scaling (1D)	7.25	6.81	0.0972	<i>0.876</i>
Trajectory Platt (5D)	9.71	9.39	0.1004	0.879
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	6.00	6.06	0.0922	<i>0.798</i>
Temperature Scaling (TS)	<i>6.52</i>	<i>6.03</i>	0.0922	<i>0.798</i>
Platt Scaling (1D)	6.59	6.02	<i>0.0923</i>	<i>0.798</i>
Trajectory Platt (5D)	7.69	7.00	0.0928	0.805

Table 19: **Temperature Transfer Robustness on pope (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	4.01	4.30	0.0903	0.876
Temperature Scaling (TS)	3.52	4.93	<i>0.0902</i>	0.876
Platt Scaling (1D)	3.97	4.64	0.0893	0.876
Trajectory Platt (5D)	<i>3.55</i>	<i>4.48</i>	0.0903	<i>0.873</i>
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	10.43	10.43	0.1108	0.872
Temperature Scaling (TS)	<i>9.42</i>	<i>9.18</i>	<i>0.1043</i>	0.872
Platt Scaling (1D)	9.20	8.91	0.1031	0.872
Trajectory Platt (5D)	10.26	10.17	0.1108	<i>0.868</i>
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	8.53	8.26	0.1098	0.852
Temperature Scaling (TS)	6.12	<i>6.16</i>	<i>0.1036</i>	0.852
Platt Scaling (1D)	<i>6.78</i>	5.66	0.1026	0.852
Trajectory Platt (5D)	7.67	6.88	0.1068	<i>0.846</i>
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	5.39	5.84	0.1081	0.855
Temperature Scaling (TS)	<i>4.31</i>	4.94	<i>0.1046</i>	0.855
Platt Scaling (1D)	4.42	<i>4.83</i>	0.1037	0.855
Trajectory Platt (5D)	2.34	3.68	0.1048	<i>0.852</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	8.46	9.36	0.1545	0.810
Temperature Scaling (TS)	8.73	<i>7.36</i>	0.1497	0.810
Platt Scaling (1D)	<i>6.48</i>	7.12	0.1473	0.810
Trajectory Platt (5D)	6.20	7.43	<i>0.1487</i>	<i>0.805</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	7.36	7.64	0.1147	0.853
Temperature Scaling (TS)	6.42	<i>6.52</i>	<i>0.1105</i>	0.853
Platt Scaling (1D)	<i>6.17</i>	6.23	0.1092	0.853
Trajectory Platt (5D)	6.01	6.53	0.1123	<i>0.849</i>

2.3 Temperature Transfer: ScienceQA

Table 20: **Temperature Transfer Robustness on scienceqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	22.16	21.93	0.2368	<i>0.807</i>
Temperature Scaling (TS)	13.10	12.68	0.1924	<i>0.807</i>
Platt Scaling (1D)	4.80	4.99	<i>0.1732</i>	<i>0.807</i>
Trajectory Platt (5D)	<i>5.17</i>	<i>6.15</i>	0.1719	0.813
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	32.97	32.97	0.3341	<i>0.786</i>
Temperature Scaling (TS)	25.55	25.61	0.2658	<i>0.786</i>
Platt Scaling (1D)	<i>25.58</i>	<i>25.08</i>	<i>0.2660</i>	<i>0.786</i>
Trajectory Platt (5D)	25.62	24.85	0.2658	0.793
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	29.33	29.02	0.2947	<i>0.797</i>
Temperature Scaling (TS)	18.72	18.70	0.2211	<i>0.797</i>
Platt Scaling (1D)	<i>16.17</i>	<i>16.77</i>	0.2082	<i>0.797</i>
Trajectory Platt (5D)	16.02	16.24	<i>0.2093</i>	0.799
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	24.79	24.47	0.2492	0.817
Temperature Scaling (TS)	15.07	14.82	0.1985	0.817
Platt Scaling (1D)	6.05	5.67	0.1726	0.817
Trajectory Platt (5D)	<i>8.14</i>	<i>7.44</i>	<i>0.1757</i>	<i>0.811</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	18.84	19.07	0.2272	0.789
Temperature Scaling (TS)	13.83	13.84	0.2068	0.789
Platt Scaling (1D)	<i>9.20</i>	<i>8.90</i>	0.1905	0.789
Trajectory Platt (5D)	8.54	7.91	<i>0.1919</i>	<i>0.783</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	25.62	25.49	0.2684	<i>0.799</i>
Temperature Scaling (TS)	17.25	17.13	0.2169	<i>0.799</i>
Platt Scaling (1D)	12.36	12.28	0.2021	<i>0.799</i>
Trajectory Platt (5D)	<i>12.70</i>	<i>12.52</i>	<i>0.2029</i>	0.800

Table 21: **Temperature Transfer Robustness on scienceqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	23.65	23.17	0.2579	<i>0.760</i>
Temperature Scaling (TS)	9.32	10.23	0.1975	<i>0.760</i>
Platt Scaling (1D)	<i>7.99</i>	<i>6.91</i>	<i>0.1916</i>	<i>0.760</i>
Trajectory Platt (5D)	6.46	5.72	0.1860	0.769
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	34.72	33.93	0.3405	<i>0.740</i>
Temperature Scaling (TS)	<i>23.17</i>	23.17	<i>0.2567</i>	<i>0.740</i>
Platt Scaling (1D)	23.39	22.36	0.2554	<i>0.740</i>
Trajectory Platt (5D)	23.04	<i>22.95</i>	0.2575	0.756
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	30.38	29.26	0.3037	<i>0.757</i>
Temperature Scaling (TS)	14.69	15.04	0.2163	<i>0.757</i>
Platt Scaling (1D)	<i>13.17</i>	12.45	<i>0.2105</i>	<i>0.757</i>
Trajectory Platt (5D)	11.66	<i>12.50</i>	0.2056	0.768
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	24.42	24.42	0.2651	<i>0.767</i>
Temperature Scaling (TS)	10.57	11.96	0.2000	<i>0.767</i>
Platt Scaling (1D)	<i>7.07</i>	<i>6.82</i>	<i>0.1902</i>	<i>0.767</i>
Trajectory Platt (5D)	6.28	5.31	0.1838	0.777
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	17.63	17.71	0.2298	<i>0.767</i>
Temperature Scaling (TS)	10.88	12.64	0.2020	<i>0.767</i>
Platt Scaling (1D)	12.93	12.97	0.2062	<i>0.767</i>
Trajectory Platt (5D)	<i>12.83</i>	<i>12.86</i>	<i>0.2033</i>	0.775
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	26.16	25.70	0.2794	<i>0.758</i>
Temperature Scaling (TS)	13.73	14.61	0.2145	<i>0.758</i>
Platt Scaling (1D)	<i>12.91</i>	<i>12.30</i>	<i>0.2108</i>	<i>0.758</i>
Trajectory Platt (5D)	12.05	11.87	0.2072	0.769

2.4 Temperature Transfer: TextVQA

Table 22: **Temperature Transfer Robustness on textvqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	10.50	11.07	0.1714	0.839
Temperature Scaling (TS)	10.38	11.14	0.1712	0.839
Platt Scaling (1D)	<i>8.47</i>	<i>8.22</i>	0.1615	0.839
Trajectory Platt (5D)	5.78	5.35	<i>0.1628</i>	<i>0.835</i>
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	27.03	25.61	0.2691	<i>0.804</i>
Temperature Scaling (TS)	26.09	24.82	0.2611	<i>0.804</i>
Platt Scaling (1D)	<i>24.52</i>	<i>22.97</i>	<i>0.2510</i>	<i>0.804</i>
Trajectory Platt (5D)	20.29	20.14	0.2314	0.808
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	15.85	14.07	0.1849	<i>0.846</i>
Temperature Scaling (TS)	13.76	13.27	0.1815	<i>0.846</i>
Platt Scaling (1D)	<i>11.19</i>	<i>10.87</i>	<i>0.1697</i>	<i>0.846</i>
Trajectory Platt (5D)	7.48	6.67	0.1575	0.852
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	<i>6.85</i>	<i>6.87</i>	0.1256	<i>0.902</i>
Temperature Scaling (TS)	7.05	7.79	0.1263	<i>0.902</i>
Platt Scaling (1D)	4.51	4.61	0.1207	<i>0.902</i>
Trajectory Platt (5D)	7.96	7.61	<i>0.1254</i>	0.904
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	<i>8.51</i>	8.63	0.1221	0.871
Temperature Scaling (TS)	8.26	<i>8.91</i>	<i>0.1233</i>	0.871
Platt Scaling (1D)	10.44	11.32	0.1313	0.871
Trajectory Platt (5D)	15.83	15.77	0.1555	0.871
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	13.75	13.25	0.1746	<i>0.852</i>
Temperature Scaling (TS)	13.11	13.19	0.1727	<i>0.852</i>
Platt Scaling (1D)	<i>11.83</i>	<i>11.60</i>	<i>0.1669</i>	<i>0.852</i>
Trajectory Platt (5D)	11.47	11.11	0.1665	0.854

Table 23: **Temperature Transfer Robustness on textvqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	32.27	31.91	0.3286	<i>0.684</i>
Temperature Scaling (TS)	18.21	18.32	0.2503	<i>0.684</i>
Platt Scaling (1D)	<i>8.95</i>	<i>10.46</i>	<i>0.2148</i>	<i>0.684</i>
Trajectory Platt (5D)	5.25	7.76	0.1948	0.719
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	56.45	56.45	0.5294	<i>0.663</i>
Temperature Scaling (TS)	<i>30.04</i>	30.04	<i>0.3007</i>	<i>0.663</i>
Platt Scaling (1D)	30.12	<i>29.53</i>	0.3187	<i>0.663</i>
Trajectory Platt (5D)	16.54	16.63	0.2147	0.761
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	48.08	48.08	0.4405	<i>0.674</i>
Temperature Scaling (TS)	25.32	25.32	0.2724	<i>0.674</i>
Platt Scaling (1D)	<i>16.80</i>	<i>17.76</i>	<i>0.2479</i>	<i>0.674</i>
Trajectory Platt (5D)	9.06	8.16	0.1763	0.774
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	37.38	37.38	0.3509	<i>0.698</i>
Temperature Scaling (TS)	22.65	22.65	0.2534	<i>0.698</i>
Platt Scaling (1D)	<i>10.82</i>	<i>10.61</i>	<i>0.1997</i>	<i>0.698</i>
Trajectory Platt (5D)	4.78	5.47	0.1538	0.802
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	31.24	31.24	0.2717	<i>0.759</i>
Temperature Scaling (TS)	28.01	28.01	0.2442	<i>0.759</i>
Platt Scaling (1D)	<i>8.41</i>	<i>7.92</i>	<i>0.1578</i>	<i>0.759</i>
Trajectory Platt (5D)	5.66	6.85	0.1333	0.810
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	41.08	41.01	0.3842	<i>0.696</i>
Temperature Scaling (TS)	24.84	24.87	0.2642	<i>0.696</i>
Platt Scaling (1D)	<i>15.02</i>	<i>15.26</i>	<i>0.2278</i>	<i>0.696</i>
Trajectory Platt (5D)	8.26	8.97	0.1746	0.773

2.5 Temperature Transfer: VizWiz-VQA

Table 24: **Temperature Transfer Robustness on vizwiz-vqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	15.97	15.97	0.2071	<i>0.793</i>
Temperature Scaling (TS)	15.09	15.09	<i>0.2065</i>	<i>0.793</i>
Platt Scaling (1D)	<i>6.64</i>	5.80	0.1828	<i>0.793</i>
Trajectory Platt (5D)	6.38	<i>6.57</i>	0.1828	0.794
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	44.03	43.59	0.4055	0.764
Temperature Scaling (TS)	41.10	40.70	0.3701	0.764
Platt Scaling (1D)	<i>34.62</i>	<i>34.81</i>	<i>0.3295</i>	0.764
Trajectory Platt (5D)	34.27	33.91	0.3284	<i>0.763</i>
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	23.72	22.45	0.2470	0.768
Temperature Scaling (TS)	20.17	19.98	0.2331	0.768
Platt Scaling (1D)	11.80	12.25	0.2090	0.768
Trajectory Platt (5D)	<i>13.33</i>	<i>13.37</i>	<i>0.2103</i>	<i>0.766</i>
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	11.58	11.10	0.1699	0.815
Temperature Scaling (TS)	10.95	10.61	0.1701	0.815
Platt Scaling (1D)	7.40	6.03	0.1588	0.815
Trajectory Platt (5D)	<i>7.80</i>	<i>8.19</i>	<i>0.1600</i>	<i>0.812</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	6.96	6.27	0.1259	0.828
Temperature Scaling (TS)	<i>8.10</i>	<i>6.71</i>	<i>0.1290</i>	0.828
Platt Scaling (1D)	9.40	9.46	0.1368	0.828
Trajectory Platt (5D)	9.60	9.67	0.1379	<i>0.826</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	20.45	19.88	0.2311	0.793
Temperature Scaling (TS)	19.08	18.62	0.2218	0.793
Platt Scaling (1D)	13.97	13.67	0.2034	0.793
Trajectory Platt (5D)	<i>14.28</i>	<i>14.34</i>	<i>0.2039</i>	<i>0.792</i>

Table 25: **Temperature Transfer Robustness on vizwiz-vqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	30.66	30.66	0.2435	0.797
Temperature Scaling (TS)	29.41	29.41	0.2328	0.797
Platt Scaling (1D)	<i>6.27</i>	7.62	<i>0.1433</i>	0.797
Trajectory Platt (5D)	6.03	<i>8.17</i>	0.1418	<i>0.795</i>
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	62.87	62.87	0.5606	<i>0.756</i>
Temperature Scaling (TS)	50.41	50.41	0.4059	<i>0.756</i>
Platt Scaling (1D)	<i>36.87</i>	<i>36.87</i>	<i>0.3228</i>	<i>0.756</i>
Trajectory Platt (5D)	32.79	32.71	0.2960	0.759
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	51.06	51.06	0.4138	<i>0.770</i>
Temperature Scaling (TS)	41.04	41.04	0.3059	<i>0.770</i>
Platt Scaling (1D)	<i>17.23</i>	<i>17.23</i>	<i>0.1811</i>	<i>0.770</i>
Trajectory Platt (5D)	13.53	14.51	0.1646	0.789
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	35.69	35.69	0.2530	0.837
Temperature Scaling (TS)	33.71	33.71	0.2374	0.837
Platt Scaling (1D)	6.25	<i>6.08</i>	<i>0.1182</i>	0.837
Trajectory Platt (5D)	<i>6.71</i>	6.06	0.1148	0.837
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	24.45	24.45	0.1509	<i>0.884</i>
Temperature Scaling (TS)	31.84	31.70	0.1971	<i>0.884</i>
Platt Scaling (1D)	<i>7.08</i>	6.42	<i>0.0916</i>	<i>0.884</i>
Trajectory Platt (5D)	6.46	<i>7.61</i>	0.0905	0.890
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	40.95	40.95	0.3244	<i>0.809</i>
Temperature Scaling (TS)	37.28	37.25	0.2758	<i>0.809</i>
Platt Scaling (1D)	<i>14.74</i>	<i>14.84</i>	<i>0.1714</i>	<i>0.809</i>
Trajectory Platt (5D)	13.11	13.81	0.1615	0.814

Table 26: **Calibration Benchmark on ai2d (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	42.00	42.00	0.4200	<i>0.614</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	8.62	12.18	0.2477	<i>0.614</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.63	7.70	<i>0.2370</i>	<i>0.614</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>8.61</i>	<i>11.15</i>	0.2289	0.661

Table 27: **Calibration Benchmark on ai2d (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	27.15	26.92	0.2887	0.746
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	13.31	13.10	0.2197	0.746
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.92	<i>8.25</i>	0.2015	0.746
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>6.27</i>	6.53	<i>0.2035</i>	<i>0.739</i>

3 Group 1: Per-Dataset In-Domain Calibration Benchmarks

Individual benchmark tables for all vision-language datasets under the 4-method, 4-metric schema.

3.1 Benchmark: AI2D

Table 28: **Calibration Benchmark on avqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	25.66	25.40	0.3194	<i>0.599</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	4.45	17.96	0.2480	<i>0.599</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	6.84	<i>17.67</i>	<i>0.2463</i>	<i>0.599</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>5.26</i>	7.45	0.2240	0.691

Table 29: **Calibration Benchmark on avqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	27.88	28.00	0.3385	<i>0.602</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	6.56	<i>13.28</i>	0.2491	<i>0.602</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	1.18	13.66	<i>0.2449</i>	<i>0.602</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>5.02</i>	6.54	0.2332	0.654

3.2 Benchmark: AVQA

Table 30: **Calibration Benchmark on chartqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	71.03	71.03	0.6151	<i>0.807</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	44.47	44.47	0.2782	<i>0.807</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	3.23	<i>3.62</i>	<i>0.0755</i>	<i>0.807</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>3.31</i>	2.88	0.0752	0.818

Table 31: **Calibration Benchmark on chartqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	71.30	71.30	0.6483	0.491
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	44.88	44.88	0.2779	0.491
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>3.06</i>	<i>4.27</i>	0.0762	<i>0.509</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	2.44	3.92	<i>0.0765</i>	0.603

3.3 Benchmark: CHARTQA

Table 32: **Calibration Benchmark on docvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	74.57	74.57	0.6430	<i>0.845</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	48.89	48.89	0.2796	<i>0.845</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>2.62</i>	<i>2.52</i>	<i>0.0405</i>	<i>0.845</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	1.87	1.81	0.0377	0.888

Table 33: **Calibration Benchmark on docvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	53.30	53.30	0.4162	<i>0.517</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	47.07	47.07	0.2584	<i>0.517</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.09	<i>3.67</i>	0.0362	<i>0.517</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>1.08</i>	2.06	<i>0.0364</i>	0.594

3.4 Benchmark: DOCVQA

Table 34: **Calibration Benchmark on gqa (M3-LLaVA 7B).** **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	32.55	32.55	0.3223	<i>0.683</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	8.41	10.85	0.2082	<i>0.683</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.90	<i>6.89</i>	<i>0.1991</i>	<i>0.683</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>7.61</i>	6.50	0.1978	0.707

Table 35: **Calibration Benchmark on gqa (MQT-LLaVA 7B).** **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	13.64	10.19	0.1865	<i>0.757</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	9.21	10.61	<i>0.1860</i>	<i>0.757</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>9.56</i>	<i>10.60</i>	0.1861	<i>0.757</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	13.81	14.66	0.1854	0.763

3.5 Benchmark: GQA

Table 36: **Calibration Benchmark on infographicvqa (M3-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	85.32	85.32	0.7761	0.333
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	54.00	54.00	0.3006	0.333
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.03</i>	<i>1.22</i>	0.0074	0.667
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.01	1.20	<i>0.0075</i>	<i>0.501</i>

Table 37: **Calibration Benchmark on infographicvqa (MQT-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	75.51	75.51	0.6535	<i>0.769</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	52.63	52.63	0.2826	<i>0.769</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.28	<i>1.11</i>	<i>0.0050</i>	<i>0.769</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.30</i>	1.08	0.0049	0.930

3.6 Benchmark: INFOGRAPHICVQA

Table 38: **Calibration Benchmark on lego-puzzles (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	73.16	73.16	0.7314	<i>0.477</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	34.71	34.71	0.3160	<i>0.477</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.71	<i>11.89</i>	<i>0.1958</i>	<i>0.477</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>4.15</i>	9.40	0.1914	0.601

Table 39: **Calibration Benchmark on lego-puzzles (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	34.60	33.80	0.3474	<i>0.510</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	23.68	23.68	0.2539	<i>0.510</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.60	<i>12.43</i>	<i>0.1968</i>	0.490
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>7.81</i>	8.67	0.1924	0.593

3.7 Benchmark: LEGOPUZZLES

Table 40: **Calibration Benchmark on mmbench (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	27.00	27.00	0.2700	<i>0.608</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>6.43</i>	<i>7.79</i>	<i>0.1956</i>	<i>0.608</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	4.60	<i>6.94</i>	0.1970	<i>0.608</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.62	6.54	0.1890	0.660

Table 41: **Calibration Benchmark on mmbench (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	15.38	15.27	0.1866	0.828
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>9.81</i>	10.24	0.1560	0.828
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	8.11	<i>8.38</i>	0.1503	0.828
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	10.42	7.51	<i>0.1543</i>	<i>0.819</i>

3.8 Benchmark: MMBENCH

Table 42: **Calibration Benchmark on mmmu (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	78.76	78.76	0.6889	0.639
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	52.21	52.21	0.2908	0.639
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>1.81</i>	3.44	<i>0.0205</i>	0.639
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	1.68	<i>3.56</i>	0.0198	<i>0.623</i>

Table 43: **Calibration Benchmark on mmmu (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	52.24	52.24	0.3669	<i>0.315</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	49.32	49.32	0.2556	<i>0.315</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.16	<i>2.14</i>	0.0115	<i>0.315</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.23</i>	2.04	<i>0.0116</i>	0.724

3.9 Benchmark: MMMU

Table 44: **Calibration Benchmark on pope (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	3.45	4.40	0.0882	<i>0.739</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	4.27	<i>4.37</i>	<i>0.0883</i>	<i>0.739</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>4.19</i>	4.32	0.0882	<i>0.739</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	4.82	6.62	0.0887	0.751

Table 45: **Calibration Benchmark on pope (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	4.01	4.30	0.0903	0.876
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	3.52	4.93	<i>0.0902</i>	0.876
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	3.97	4.64	0.0893	0.876
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>3.55</i>	<i>4.48</i>	0.0903	<i>0.873</i>

3.10 Benchmark: POPE

Table 46: **Calibration Benchmark on scienceqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	22.16	21.93	0.2368	<i>0.807</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	13.10	12.68	0.1924	<i>0.807</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	4.80	4.99	<i>0.1732</i>	<i>0.807</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>5.17</i>	<i>6.15</i>	0.1719	0.813

Table 47: **Calibration Benchmark on scienceqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	23.65	23.17	0.2579	<i>0.760</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	9.32	10.23	0.1975	<i>0.760</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>7.99</i>	<i>6.91</i>	<i>0.1916</i>	<i>0.760</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.46	5.72	0.1860	0.769

3.11 Benchmark: SCIENCEQA

Table 48: **Calibration Benchmark on seedbench (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	32.25	32.25	0.3225	<i>0.483</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>0.66</i>	<i>5.25</i>	0.2185	<i>0.483</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.48	5.20	<i>0.2184</i>	<i>0.483</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	3.94	7.68	0.2071	0.642

Table 49: **Calibration Benchmark on seedbench (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	18.56	17.74	0.2297	<i>0.733</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>6.68</i>	10.44	0.2015	<i>0.733</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	9.51	8.62	<i>0.1986</i>	<i>0.733</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.66	<i>9.67</i>	0.1957	0.741

3.12 Benchmark: SEEDBENCH

Table 50: **Calibration Benchmark on textvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	10.50	11.07	0.1714	0.839
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	10.38	11.14	0.1712	0.839
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>8.47</i>	<i>8.22</i>	0.1615	0.839
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.78	5.35	<i>0.1628</i>	<i>0.835</i>

Table 51: **Calibration Benchmark on textvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	32.27	31.91	0.3286	<i>0.684</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	18.21	18.32	0.2503	<i>0.684</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>8.95</i>	<i>10.46</i>	<i>0.2148</i>	<i>0.684</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.25	7.76	0.1948	0.719

3.13 Benchmark: TEXTVQA

Table 52: **Calibration Benchmark on vizwiz-vqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	15.97	15.97	0.2071	<i>0.793</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	15.09	15.09	<i>0.2065</i>	<i>0.793</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>6.64</i>	5.80	0.1828	<i>0.793</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.38	<i>6.57</i>	0.1828	0.794

Table 53: **Calibration Benchmark on vizwiz-vqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	30.66	30.66	0.2435	0.797
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	29.41	29.41	0.2328	0.797
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>6.27</i>	7.62	<i>0.1433</i>	0.797
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.03	<i>8.17</i>	0.1418	<i>0.795</i>

3.14 Benchmark: VIZWIZVQA

Table 54: **Calibration Benchmark on vllm-safety (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	<i>11.59</i>	<i>10.50</i>	<i>0.1732</i>	0.814
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	16.98	16.80	0.1812	0.814
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	14.76	14.04	0.1798	0.814
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.87	6.88	0.1709	<i>0.773</i>

Table 55: **Calibration Benchmark on vllm-safety (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	<i>9.63</i>	<i>9.74</i>	0.1966	0.763
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	14.20	14.42	0.2054	0.763
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	11.34	11.29	<i>0.1937</i>	0.763
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.68	6.58	0.1921	<i>0.733</i>

3.15 Benchmark: VLLM-SAFETY

Table 56: **Calibration Benchmark on vqav2 (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	8.64	<i>7.23</i>	<i>0.1387</i>	<i>0.772</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	9.40	8.58	0.1417	<i>0.772</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.52	<i>6.97</i>	0.1391	<i>0.772</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>6.86</i>	6.50	0.1380	0.788

Table 57: **Calibration Benchmark on vqav2 (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	<i>10.49</i>	<i>10.31</i>	0.1713	<i>0.641</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	13.87	13.65	0.1789	<i>0.641</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	10.54	10.94	<i>0.1712</i>	<i>0.641</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	7.16	7.64	0.1629	0.739

3.16 Benchmark: VQAV2

Table 58: **Combinatorial 5D Trajectory Feature Cardinality Progression** ($k \in \{1 \dots 5\}$). Macro-averaged out-of-fold calibration error and discrimination across all 14 vision-language benchmarks for M3-LLaVA (7B) and MQT-LLaVA (7B). Evaluates all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets. Reported in pure decimal format. **Bold**: Rank 1, *Italic*: Rank 2 within each architecture. Note: Subsets lacking the base logit anchor x_1 achieve artificially low quantile Ada-ECE via class-prior probability clustering, but suffer severe AUROC collapse (≤ 0.64 vs. $0.71+$ for x_1 -anchored models); joint calibration and discrimination require the full 5D representation.

Cardinality (k)	Subsets ($\binom{5}{k}$)	Optimal Formula	Macro Ada-ECE (Best) $\downarrow$	Macro Ada-ECE (Mean $\pm$)
M3-LLaVA (7B)				
$k = 1$	5	x_5	0.0270	0.0338 ± 0.0082
$k = 2$	10	$x_2 + x_4$	<i>0.0261</i>	0.0333 ± 0.0043
$k = 3$	10	$x_2 + x_4 + x_5$	0.0256	0.0317 ± 0.0031
$k = 4$	5	$x_1 + x_3 + x_4 + x_5$	0.0288	0.0297 ± 0.0007
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0274	0.0274 ± 0.0000
MQT-LLaVA (7B)				
$k = 1$	5	x_5	0.0280	0.0355 ± 0.0079
$k = 2$	10	$x_2 + x_4$	<i>0.0261</i>	0.0358 ± 0.0069
$k = 3$	10	$x_2 + x_4 + x_5$	0.0255	0.0326 ± 0.0048
$k = 4$	5	$x_1 + x_2 + x_3 + x_4$	0.0293	0.0317 ± 0.0019
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0320	0.0320 ± 0.0000

4 Group 8: Combinatorial 5D Trajectory Feature Ablation Study

Exhaustive 5-fold cross-validation combinatorial ablation across all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets of canonical 5D trajectory features across all 14 vision-language benchmarks on M3-LLaVA (7B) and MQT-LLaVA (7B).

4.1 Table 4: Macro Cardinality Progression ($k \in \{1 \dots 5\}$)

Table 59: **Leave-One-Out (LOO) Feature Degradation Sensitivity across 14 Benchmarks.** Marginal change in calibration error ($\Delta\text{Ada-ECE} = \text{Ada-ECE}_{-j} - \text{Ada-ECE}_{\text{Full 5D}}$) and discrimination ($\Delta\text{AUROC} = \text{AUROC}_{-j} - \text{AUROC}_{\text{Full 5D}}$) when dropping individual feature x_j from Trajectory Platt (5D). Positive $\Delta\text{Ada-ECE}$ and negative ΔAUROC indicate feature necessity. **Bold:** Rank 1, *Italic:* Rank 2 degradation impact.

Dropped Feature	Functional Role	M3-LLaVA (7B)		MQT-I
		Δ Ada-ECE	Δ AUROC	Δ Ada-EC
x_1 (Final Logit)	Base Fine-Scale Log-Odds Anchor ($\ell_{576/256}$)	+0.0026	-0.091	+0.0025
x_2 (Discrete Answer Stability)	Categorical Scale Consistency	+0.0014	+0.001	-0.0008
x_3 (Scale Entropy Slope)	Predictive Entropy Decay Dynamics	<i>+0.0029</i>	<i>-0.004</i>	<i>+0.0013</i>
x_4 (Answer Flip Frequency)	Argmax Scale Transition Rate	+0.0015	-0.000	-0.0015
x_5 (Monotonicity Count)	Directional Monotonicity Trend	+0.0031	-0.003	-0.0027

4.2 Table 5: Leave-One-Out (LOO) Feature Degradation Sensitivity

Table 60: **Exhaustive 14-Benchmark Progression Grid: Adaptive ECE ($\downarrow$)**. Per-dataset comparison across 1D Platt ($\{x_1\}$), Optimal 2D, Optimal 3D, Optimal 4D, and Full Trajectory Platt 5D. Option A ranking: **Bold** Rank 1, *Italic* Rank 2 across each benchmark row.

Benchmark Dataset	1D Platt ($\{x_1\}$)	Best-2D	Best-3D	Best-4D	Full 5D (Ours)
M3-LLaVA (7B)					
AI2D	0.0418	0.0311	<i>0.0335</i>	0.0440	0.0620
ChartQA	0.0149	0.0124	<i>0.0129</i>	0.0139	0.0153
DocVQA	0.0115	<i>0.0085</i>	0.0088	0.0083	0.0121
GQA	0.0411	<i>0.0255</i>	0.0243	0.0261	0.0332
InfoVQA	0.0053	0.0044	<i>0.0047</i>	0.0056	0.0074
LegoPuzzles	0.0949	0.0459	<i>0.0357</i>	0.0413	0.0345
MMBench	0.0419	0.0256	<i>0.0271</i>	0.0355	0.0420
MMMU	0.0131	0.0092	<i>0.0098</i>	0.0122	0.0155
POPE	0.0137	<i>0.0116</i>	0.0104	0.0135	0.0185
ScienceQA	0.0407	0.0208	<i>0.0292</i>	0.0370	0.0351
SEED-Bench	0.0274	<i>0.0259</i>	0.0269	0.0263	0.0230
TextVQA	0.0432	<i>0.0226</i>	0.0166	0.0359	0.0376
VizWiz-VQA	0.0259	0.0172	0.0202	<i>0.0188</i>	0.0204
VQAv2	0.0764	0.0204	<i>0.0206</i>	0.0238	0.0272
Macro Mean	0.0351	0.0201	0.0201	<i>0.0244</i>	0.0274
MQT-LLaVA (7B)					
AI2D	0.0362	0.0152	<i>0.0176</i>	0.0294	0.0371
ChartQA	0.0454	0.0165	<i>0.0123</i>	0.0122	0.0179
DocVQA	0.0231	<i>0.0083</i>	0.0077	<i>0.0083</i>	0.0118
GQA	0.0525	0.0287	<i>0.0335</i>	0.0353	0.0544
InfoVQA	0.0065	0.0063	0.0054	0.0046	<i>0.0051</i>
LegoPuzzles	0.0899	0.0395	0.0303	<i>0.0383</i>	0.0520
MMBench	0.0407	0.0321	<i>0.0391</i>	0.0453	0.0490
MMMU	0.0128	0.0098	0.0079	<i>0.0090</i>	0.0116
POPE	0.0236	0.0198	<i>0.0175</i>	0.0171	0.0210
ScienceQA	0.0511	0.0329	0.0282	<i>0.0310</i>	0.0404
SEED-Bench	0.0303	0.0227	<i>0.0299</i>	0.0338	0.0530
TextVQA	0.0997	0.0223	<i>0.0225</i>	0.0386	0.0345
VizWiz-VQA	0.0446	0.0092	<i>0.0237</i>	0.0302	0.0305
VQAv2	0.0880	0.0232	0.0256	<i>0.0255</i>	0.0296
Macro Mean	0.0460	0.0205	<i>0.0215</i>	0.0256	0.0320

4.3 Table 6: Exhaustive 14-Benchmark Adaptive ECE Progression Grid

Table 61: **Combinatorial 5D Trajectory Feature Cardinality Progression on TextVQA** ($k \in \{1 \dots 5\}$). Out-of-fold calibration error and discrimination across feature cardinality levels for M3-LLaVA (7B) and MQT-LLaVA (7B). Evaluated across all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets. Reported in pure decimal format. **Bold**: Rank 1, *Italic*: Rank 2 within each architecture. Note: Subsets lacking the base logit anchor x_1 achieve artificially low quantile Ada-ECE via class-prior probability clustering, but suffer severe AUROC collapse and degraded Brier scores; joint calibration and discrimination require the full 5D representation.

Cardinality (k)	Subsets	Optimal Formula	Ada-ECE ↓	Mean ± Std	AUROC ↑	Brier ↓
M3-LLaVA (7B)						
$k = 1$	5	x_5	0.0297	0.0504 ± 0.0192	0.583	0.2371
$k = 2$	10	$x_4 + x_5$	<i>0.0226</i>	0.0501 ± 0.0144	<i>0.731</i>	0.2039
$k = 3$	10	$x_2 + x_4 + x_5$	0.0166	0.0425 ± 0.0111	0.730	0.2040
$k = 4$	5	$x_1 + x_3 + x_4 + x_5$	0.0359	0.0402 ± 0.0027	0.842	<i>0.1595</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0376	0.0376 ± 0.0000	0.842	0.1593
MQT-LLaVA (7B)						
$k = 1$	5	x_5	0.0223	0.0527 ± 0.0307	0.607	0.2197
$k = 2$	10	$x_2 + x_4$	0.0223	0.0456 ± 0.0149	0.633	0.2163
$k = 3$	10	$x_2 + x_4 + x_5$	<i>0.0225</i>	0.0427 ± 0.0115	0.682	0.2070
$k = 4$	5	$x_1 + x_2 + x_4 + x_5$	0.0386	0.0448 ± 0.0054	<i>0.738</i>	<i>0.1926</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0345	0.0345 ± 0.0000	0.747	0.1851

4.4 Table 7: Cardinality Progression on TextVQA

Table 62: **Combinatorial 5D Trajectory Feature Cardinality Progression on VQAv2** ($k \in \{1 \dots 5\}$). Out-of-fold calibration error and discrimination across feature cardinality levels for M3-LLaVA (7B) and MQT-LLaVA (7B). Evaluated across all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets. Reported in pure decimal format. **Bold**: Rank 1, *Italic*: Rank 2 within each architecture. Note: Subsets lacking the base logit anchor x_1 achieve artificially low quantile Ada-ECE via class-prior probability clustering, but suffer severe AUROC collapse and degraded Brier scores; joint calibration and discrimination require the full 5D representation.

Cardinality (k)	Subsets	Optimal Formula	Ada-ECE ↓	Mean ± Std	AUROC ↑	Brier ↓
M3-LLaVA (7B)						
$k = 1$	5	x_5	0.0271	0.0411 ± 0.0202	0.580	0.1634
$k = 2$	10	$x_4 + x_5$	0.0204	0.0408 ± 0.0169	0.710	0.1481
$k = 3$	10	$x_2 + x_4 + x_5$	<i>0.0206</i>	0.0361 ± 0.0111	0.714	0.1464
$k = 4$	5	$x_2 + x_3 + x_4 + x_5$	0.0238	0.0316 ± 0.0059	<i>0.752</i>	<i>0.1431</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0272	0.0272 ± 0.0000	0.788	0.1362
MQT-LLaVA (7B)						
$k = 1$	5	x_4	0.0259	0.0430 ± 0.0257	0.661	0.1633
$k = 2$	10	$x_4 + x_5$	0.0232	0.0385 ± 0.0140	0.703	0.1611
$k = 3$	10	$x_2 + x_4 + x_5$	0.0256	0.0327 ± 0.0062	0.700	0.1614
$k = 4$	5	$x_1 + x_2 + x_4 + x_5$	<i>0.0255</i>	0.0285 ± 0.0019	<i>0.729</i>	<i>0.1558</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0296	0.0296 ± 0.0000	0.771	0.1511

4.5 Table 8: Cardinality Progression on VQAv2

Table 63: **Combinatorial 5D Trajectory Feature Cardinality Progression on LegoPuzzles** ($k \in \{1 \dots 5\}$). Out-of-fold calibration error and discrimination across feature cardinality levels for M3-LLaVA (7B) and MQT-LLaVA (7B). Evaluated across all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets. Reported in pure decimal format. **Bold**: Rank 1, *Italic*: Rank 2 within each architecture. Note: Subsets lacking the base logit anchor x_1 achieve artificially low quantile Ada-ECE via class-prior probability clustering, but suffer severe AUROC collapse and degraded Brier scores; joint calibration and discrimination require the full 5D representation.

Cardinality (k)	Subsets	Optimal Formula	Ada-ECE $\downarrow$	Mean $\pm$ Std	AUROC $\uparrow$	Brier $\downarrow$
M3-LLaVA (7B)						
$k = 1$	5	x_5	0.0411	0.0581 ± 0.0216	0.516	0.1971
$k = 2$	10	$x_2 + x_4$	0.0459	0.0589 ± 0.0151	0.494	0.1981
$k = 3$	10	$x_3 + x_4 + x_5$	<i>0.0357</i>	0.0568 ± 0.0186	0.589	0.1927
$k = 4$	5	$x_1 + x_2 + x_3 + x_5$	0.0413	0.0495 ± 0.0106	<i>0.583</i>	0.1927
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0345	0.0345 ± 0.0000	0.579	<i>0.1932</i>
MQT-LLaVA (7B)						
$k = 1$	5	x_2	<i>0.0337</i>	0.0549 ± 0.0212	0.485	0.1982
$k = 2$	10	$x_2 + x_4$	0.0395	0.0623 ± 0.0260	0.485	0.2002
$k = 3$	10	$x_1 + x_3 + x_4$	0.0303	0.0535 ± 0.0224	0.613	0.1913
$k = 4$	5	$x_1 + x_2 + x_3 + x_4$	0.0383	0.0536 ± 0.0172	0.607	0.1934
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0520	0.0520 ± 0.0000	<i>0.611</i>	<i>0.1929</i>

4.6 Table 9: Cardinality Progression on LegoPuzzles